More information

Modelfile Reference

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A Modelfile is the blueprint to create and share customized models using Ollama.

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```
# comment
INSTRUCTION arguments
```

Instruction	Description
`FROM` (required)	Defines the base model to use.
`PARAMETER`	Sets the parameters for how Ollama will run the model.
`TEMPLATE`	The full prompt template to be sent to the model.
`SYSTEM`	Specifies the system message that will be set in the template.
`ADAPTER`	Defines the (Q)LoRA adapters to apply to the model.
`LICENSE`	Specifies the legal license.
`MESSAGE`	Specify message history.
`REQUIRES`	Specify the minimum version of Ollama required by the model.

Examples

Basic `Modelfile`

An example of a `Modelfile` creating a mario blueprint:

```
FROM llama3.2
# sets the temperature to 1 [higher is more creative, lower is more coher
PARAMETER temperature 1
# sets the context window size to 4096, this controls how many tokens the
PARAMETER num_ctx 4096

# sets a custom system message to specify the behavior of the chat assist
SYSTEM You are Mario from super mario bros, acting as an assistant.
```

To use this:



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4. Start using the model!

To view the Modelfile of a given model, use the `ollama show --modelfile` command.

```
ollama show --modelfile llama3.2
```

```
# Modelfile generated by "ollama show"
# To build a new Modelfile based on this one, replace the FROM line with:
# FROM llama3.2:latest
FROM /Users/pdevine/.ollama/models/blobs/sha256-00e1317cbf74d901080d7100f
TEMPLATE """{{ if .System }}<|start_header_id|>system<|end_header_id|>

{{ .System }}<|eot_id|>{{ end }}{{ if .Prompt }}<|start_header_id|>user<|

{{ .Prompt }}<|eot_id|>{{ end }}<|start_header_id|>assistant<|end_header_

{{ .Response }}<|eot_id|>"""
PARAMETER stop "<|start_header_id|>"
PARAMETER stop "<|end_header_id|>"
PARAMETER stop "<|eot_id|>"
PARAMETER stop "<|reserved_special_token"
```

Instructions

FROM (Required)

The `FROM` instruction defines the base model to use when creating a model.

```
FROM <model name>:<tag>
```

Build from existing model



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Base Models

A list of available base models

Base Models

Additional models can be found at

Build from a Safetensors model

```
FROM <model directory>
```

The model directory should contain the Safetensors weights for a supported architecture.

Currently supported model architectures:

- Llama (including Llama 2, Llama 3, Llama 3.1, and Llama 3.2)
- Mistral (including Mistral 1, Mistral 2, and Mixtral)
- Gemma (including Gemma 1 and Gemma 2)
- Phi3

Build from a GGUF file

```
FROM ./ollama-model.gguf
```

The GGUF file location should be specified as an absolute path or relative to the `Modelfile` location.

PARAMETER

Valid Parameters and Values

Parameter	Description	Value Type	Example Usage
num_ctx	Sets the size of the context window used to generate the next token. (Default: 2048)	int	num_ctx 4096
repeat_last_n	Sets how far back for the model to look back to prevent repetition. (Default: 64, 0 = disabled, -1 = num_ctx)	int	repeat_last_n 64
repeat_penalty	Sets how strongly to penalize repetitions. A higher value (e.g., 1.5) will penalize repetitions more strongly, while a lower value (e.g., 0.9) will be more lenient. (Default: 1.0)	float	repeat_penalty 1.0
presence_penalty	Penalizes tokens that have already appeared in the generated text to reduce repetition. (Default: 0.0)	float	presence_penalty 1.5
frequency_penalty	Penalizes tokens based on how often they have appeared in the generated text. (Default: 0.0)	float	frequency_penalty 1.0
temperature	The temperature of the model. Increasing the temperature will make the model answer more	float	temperature 0.7

	for generation. Setting this to a specific number will make the model generate the same text for the same prompt. (Default: 0)		
stop	Sets the stop sequences to use. When this pattern is encountered the LLM will stop generating text and return. Multiple stop patterns may be set by specifying multiple separate `stop` parameters in a modelfile.	string	stop “AI assistant:”
num_predict	Maximum number of tokens to predict when generating text. (Default: -1, infinite generation)	int	num_predict 42
top_k	Reduces the probability of generating nonsense. A higher value (e.g. 100) will give more diverse answers, while a lower value (e.g. 10) will be more conservative. (Default: 40)	int	top_k 40
top_p	Works together with top-k. A higher value (e.g., 0.95) will lead to more diverse text, while a lower value (e.g., 0.5) will generate more focused and	float	top_p 0.9

	ensure a balance of quality and variety. The parameter _p represents the minimum probability for a token to be considered, relative to the probability of the most likely token. For example, with *p*=0.05 and the most likely token having a probability of 0.9, logits with a value less than 0.045 are filtered out. (Default: 0.0)

TEMPLATE

`TEMPLATE` of the full prompt template to be passed into the model. It may include (optionally) a system message, a user's message and the response from the model. Note: syntax may be model specific. Templates use Go **template syntax**.

Template Variables

Variable	Description
`{{ .System }}`	The system message used to specify custom behavior.
`{{ .Prompt }}`	The user prompt message.
`{{ .Response }}`	The response from the model. When generating a response, text after this variable is omitted.

```
{{ end }}<|im_start|>assistant
"""
```

SYSTEM

The `SYSTEM` instruction specifies the system message to be used in the template, if applicable.

```
SYSTEM """<system message>"""
```

ADAPTER

The `ADAPTER` instruction specifies a fine tuned LoRA adapter that should apply to the base model. The value of the adapter should be an absolute path or a path relative to the Modelfile. The base model should be specified with a `FROM` instruction. If the base model is not the same as the base model that the adapter was tuned from the behaviour will be erratic.

Safetensor adapter

```
ADAPTER <path to safetensor adapter>
```

Currently supported Safetensor adapters:

- Llama (including Llama 2, Llama 3, and Llama 3.1)
- Mistral (including Mistral 1, Mistral 2, and Mixtral)
- Gemma (including Gemma 1 and Gemma 2)

GGUF adapter



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LICENSE

The `LICENSE` instruction allows you to specify the legal license under which the model used with this Modelfile is shared or distributed.

```
LICENSE """
<license text>
"""
```

MESSAGE

The `MESSAGE` instruction allows you to specify a message history for the model to use when responding. Use multiple iterations of the MESSAGE command to build up a conversation which will guide the model to answer in a similar way.

```
MESSAGE <role> <message>
```

Valid roles

Role	Description
system	Alternate way of providing the SYSTEM message for the model.
user	An example message of what the user could have asked.
assistant	An example message of how the model should respond.

Example conversation

```
MESSAGE user is Ontario in Canada?
MESSAGE assistant yes
```

REQUIRES

The `REQUIRES` instruction allows you to specify the minimum version of Ollama required by the model.

```
REQUIRES <version>
```

The version should be a valid Ollama version (e.g. 0.14.0).

Notes

the `Modelfile` **is not case sensitive**. In the examples, uppercase instructions are used to make it easier to distinguish it from arguments.

Instructions can be in any order. In the examples, the `FROM` instruction is first to keep it easily readable.

Previous Context length Next